

Poisson $\quad P[N(t)=n] = \dfrac{(\lambda t)^n}{\lfloor n} e^{-\lambda t}$

$$P[\overbrace{(N(b)-N(a))}^{A}=0] = P[N(b-a)=0] = \frac{[\lambda(b-a)]^0}{\lfloor 0} e^{-\lambda(b-a)} = e^{-\lambda(b-a)}$$

$$P[\overbrace{N(d)-N(c)=0}^{B}] = e^{-\lambda(d-c)}$$

$$P[A \text{ and } B] = P[A]\cdot P[B] = e^{-\lambda(b-a)} \cdot e^{-\lambda(d-c)}$$

$$= e^{-\lambda(\alpha+\beta)} \quad \text{—— (1)}$$

$$P[N(\alpha+\beta)=0] = \frac{[\lambda(\alpha+\beta)]^0 e^{-\lambda(\alpha+\beta)}}{0!} = e^{-\lambda(\alpha+\beta)} \quad \text{—— (2)}$$

(1) and (2) $\Rightarrow$ probability [no events in disjoint intervals]

= probability [no events in the duration of all disjoint intervals]

compare with the proof of last couple of pages in lecture notes of Poisson Arrival Process



